

The Islamia University of Bahawalpur



(Project Proposal)

AI-Powered Resume Screening Tool

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University Project Proposal

AI-Powered Resume Screening Tool

1. Introduction

In today's competitive job market, companies receive hundreds or even thousands of applications for a single position. Human resource professionals spend countless hours manually reviewing resumes on their computers, a process that is not only time-consuming but also prone to human error and unconscious bias. This project aims to develop a mobile-based intelligent resume screening application that leverages artificial intelligence and natural language processing to automate the initial candidate evaluation process. The app will allow recruiters to screen candidates anytime, anywhere, directly from their smartphones or tablets.

2. Problem Statement

The traditional resume screening process faces several critical challenges that impact hiring quality and efficiency.

- Recruiters spend approximately 20 to 30 hours manually screening 100 resumes on desktop computers
 - This manual process leads to fatigue, causing qualified candidates to be overlooked
 - Recruiters spend an average of only 7 seconds reviewing each resume before making a decision
 - Unconscious bias affects hiring decisions based on name, gender, educational background, and formatting style
 - Different recruiters have different evaluation criteria leading to inconsistency
 - Large organizations receive millions of applications annually making desktop-only screening inadequate
 - Many qualified candidates are rejected simply because their resume was not seen
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3. Project Objectives

Primary Objectives

- To design and develop a mobile application for resume screening on Android and iOS platforms
- To implement natural language processing for accurate information extraction from resumes
- To create a candidate ranking algorithm that scores applicants based on job relevance
- To reduce screening time by 80 percent compared to manual desktop-based screening
- To provide mobile-first experience with offline capabilities for screening without internet
- To deliver a user-friendly interface optimized for touch interaction on small screens

Secondary Objectives

- To implement camera integration for scanning physical resumes
 - To provide push notifications for screening results and candidate updates
 - To enable cloud synchronization across multiple devices
 - To implement biometric authentication for secure access
 - To generate mobile-friendly reports that can be shared instantly
 - To support voice input for job description entry
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4. Proposed System

The proposed system is a mobile application consisting of three main components working together to provide seamless resume screening on smartphones and tablets.

Resume Scanning and Parsing Component

Recruiters can upload resumes through multiple methods on their mobile devices. The app allows uploading from device storage, cloud storage services, and email attachments.

Job Description Manager

Recruiters can enter job descriptions through typing or voice input on their mobile devices.

Matching and Ranking Engine

This component runs both on-device and in the cloud depending on processing needs. It compares each candidate's profile against job requirements

Mobile Dashboard

The dashboard displays active job postings and recent screening results. Recruiters can view ranked candidate lists optimized for mobile viewing

5. Key Features

- Mobile-First Design
 - Multiple Resume Upload Options
 - Camera Integration
 - Voice Input for Job Descriptions
 - Offline Mode
 - Push Notifications
 - Biometric Authentication
 - Instant Sharing
 - Cloud Synchronization
 - Smart Recommendations
 - Analytics Dashboard
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6. Technologies Used

Frontend

- HTML
- CSS
- Tailwind CSS
- JavaScript
- React.js

Backend

- Node.js
- Express.js

Database

- MongoDB / MySQL

Other Tools

- GitHub
 - VS Code
 - Postman
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7. System Architecture

Mobile Application Layer

The mobile app runs on user devices with both online and offline capabilities. User interface components include screens for authentication, job management, resume upload, candidate viewing, and analytics. Local storage caches recent data for offline access. Background services handle synchronization when connectivity returns. Native modules access device features like camera, biometrics, and notifications.

Cloud Services Layer

Firebase provides authentication, real-time database, and cloud storage. Cloud Functions run serverless code for heavy processing tasks. When a user uploads resumes, Cloud Functions trigger parsing and matching algorithms. Results are stored and

synchronized back to the mobile app. Push notifications are sent through Firebase Cloud Messaging.

Data Storage Layer

Firestore stores structured data including user profiles, job descriptions, and candidate information. Cloud Storage stores original resume files and scanned images. IndexedDB on the device maintains offline cache. Local storage saves user preferences and recent searches.

8. Methodology

The project will follow Agile methodology with two-week sprints for mobile development.

Sprint 1: Project Setup and Research

Sprint 2: Authentication and User Profiles

Sprint 3: Resume Upload and Scanning

Sprint 4: OCR and Text Extraction

Sprint 5: Resume Parsing Engine

Sprint 6: Job Description Management

Sprint 7: Matching and Ranking

Sprint 8: Offline Mode and Synchronization

Sprint 9: Push Notifications and Sharing

Sprint 10: Analytics and Reporting

Sprint 11: Testing and Optimization

Sprint 12: Deployment and Documentation

9. Expected Outcomes

Quantitative Outcomes

- Mobile app reduces resume screening time by 80 percent
 - Processing speed reaches 100 resumes per hour on mobile
 - Camera scanning accuracy reaches 85 percent for printed resumes
 - Recruiters can screen candidates during commutes and travel
 - Paper resumes can be scanned instantly without computers
 - Voice input makes job description entry faster and easier
 - Biometric authentication provides quick and secure access
 - Instant sharing enables faster collaboration with hiring teams
 - Fully functional mobile app for iOS and Android
 - Cross-platform codebase with 80 percent code reuse
 - Seamless cloud synchronization across devices
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10. Future Enhancements

Apple Watch and Wearable Integration

- Quick candidate previews on smartwatches
- Screening alerts and notifications on wearables
- Voice commands for basic screening tasks
- Haptic feedback for important notifications

Tablet Optimized Interface

- Split-screen view for candidate comparison
 - Landscape mode for detailed reviews
 - Multi-window support for productivity
 - Stylus support for notes and annotations
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11. Budget Estimate

Development Resources

Cloud infrastructure for hosting and processing will cost approximately 50,000 PKR. This includes virtual servers, database hosting, storage, and content delivery networks.

Development tools and software licenses will require 10,000 PKR for IDEs, design tools, and testing services.

Data Collection and Annotation

Collecting and labeling training data will cost 15,000 PKR. This includes acquiring diverse resumes and paying for expert annotation services to create high-quality training datasets.

Personnel Costs

While this is an academic project, if considering commercial development, personnel costs would include project manager, machine learning engineer, frontend developer, backend developer, and quality assurance tester.

Infrastructure and Deployment

Domain registration and SSL certificate costs 5,000 PKR annually. Deployment and configuration services require 5,000 PKR. Monitoring and maintenance tools cost 5,000 PKR.

Documentation and Testing

Documentation and presentation materials cost 5,000 PKR. Testing services and user acceptance testing require 5,000 PKR. Contingency for unexpected costs is set at 10,000

12. Risks and Mitigation

Risk 1: Resume Format Variability

Risk 2: Accuracy Concerns

Risk 3: Data Privacy and Security

Risk 4: Scalability Issues.

Risk 5: Technology Integration Complexity

13. Conclusion

The AI-Powered Resume Screening Tool addresses a critical need in modern recruitment. Manual resume screening is inefficient, inconsistent, and prone to bias. Organizations struggle to handle high application volumes while ensuring they identify the best candidates.

This project proposes an intelligent solution that leverages artificial intelligence and natural language processing to automate initial candidate evaluation. The system extracts relevant information from resumes, analyzes job requirements, and ranks candidates based on qualifications. It provides transparency through explainable results and helps reduce bias through standardized evaluation.